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Exercise list: L09 — Multivariate time series (Class 9): textbook exercises

Instructions. These are the *Practice* items of Class 9 in `Map/class-plan.md`: Tsay 8.1, 8.2, 8.5, 8.7; Tsay 9.7; Tsay 10.1–10.6; Greenberg 10.4. Every question opens with a source line, and the statement is reproduced verbatim from the book PDF. Data exercises use Tsay's data files (`Listas/data/`, provenance in `Listas/data/README.md`) and are solved in Python (NumPy, SciPy, `statsmodels`); the multivariate GARCH models are estimated by Gaussian maximum likelihood written in NumPy and `scipy.optimize`. `Listas/L09-book-exercises-check.py` reproduces and asserts every number in the solutions. Show your steps.

Block 1 — Vector autoregressions, IRFs, forecasts and cointegration *(Tsay Ch. 8 • Questions 1–4)*

Question 1. Cross-correlation matrices and the multivariate portmanteau test (Tsay 8.1).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 8, Exercise 8.1, printed p. 462. Reproduced verbatim.

> 8.1. Consider the monthly log stock returns, in percentages and including dividends, of Merck & Company, Johnson & Johnson, General Electric, General Motors, Ford Motor Company, and value-weighted index from January 1960 to December 2008; see the file `m-mrk2vw.txt`.
>
> (a) Compute the sample mean, covariance matrix, and correlation matrix of the data.
>
> (b) Test the hypothesis $H_0 : \boldsymbol{\rho}_1 = \cdots = \boldsymbol{\rho}_6 = \mathbf{0}$, where $\boldsymbol{\rho}_i$ is the lag-i cross-correlation matrix of the data. Draw conclusions based on the 5% significance level.
>
> (c) Is there any lead–lag relationship among the six return series?

Solution:

Question 2. Bivariate AR and MA models for interest-rate changes; structural form (Tsay 8.2).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 8, Exercise 8.2, printed pp. 462–463. Reproduced verbatim.

> 8.2. The Federal Reserve Bank of St. Louis publishes selected interest rates and U.S. financial data on its website: `http://research.stlouisfed.org/fred2/`. Consider the monthly 1-year and 10-year Treasury constant maturity rates from April 1953 to October 2009 for 679 observations; see the file `m-gs1n10.txt`. The rates are in percentages.
>
> (a) Let $c_t = r_t - r_{t-1}$ be the change series of the monthly interest rate r_t. Build a bivariate autoregressive model for the two change series. Discuss the implications of the model. Transform the model into a structural form.
>
> (b) Build a bivariate moving-average model for the two change series. Discuss the implications of the model and compare it with the bivariate AR model built earlier.

Solution:

Question 3. A seasonal bivariate AR(4) model (Tsay 8.5).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 8, Exercise 8.5, printed p. 463. Reproduced verbatim.

> 8.5. The bivariate AR(4) model $\boldsymbol{x}_t - \boldsymbol{\Phi}_4 \boldsymbol{x}_{t-4} = \boldsymbol{\phi}_0 + \boldsymbol{a}_t$ is a special seasonal model with periodicity 4, where $\{\boldsymbol{a}_t\}$ is a sequence of independent and identically distributed normal random vectors with mean zero and covariance matrix $\boldsymbol{\Sigma}$. Such a seasonal model may be useful in studying quarterly earnings of a company. (a) Assume that $\boldsymbol{x}_t$ is weakly stationary. Derive the mean vector and covariance matrix of $\boldsymbol{x}_t$. (b) Derive the necessary and sufficient condition of weak stationarity for $\boldsymbol{x}_t$. (c) Show that $\boldsymbol{\Gamma}_\ell = \boldsymbol{\Phi}_4 \boldsymbol{\Gamma}_{\ell-4}$ for $\ell > 0$, where $\boldsymbol{\Gamma}_\ell$ is the lag-ℓ autocovariance matrix of $\boldsymbol{x}_t$.

Solution:

Question 4. VAR, impulse responses, forecasts, cointegration and ECM for 1- and 3-year rates (Tsay 8.7).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 8, Exercise 8.7, printed pp. 463–464. Reproduced verbatim.

> 8.7. Consider the monthly U.S. 1-year and 3-year Treasury constant maturity rates from April 1953 to March 2004. The data can be obtained from the Federal Reserve Bank of St. Louis or from the file `m-gs1n3-5304.txt` (1-year, 3-year, dates). See also Example 8.6, which uses a shorter data span. Here we use the interest rates directly without the log transformation and define $\boldsymbol{x}_t = (x_{1t}, x_{2t})'$, where x_{1t} is the 1-year maturity rate and x_{2t} is the 3-year maturity rate.
>
> (a) Identify a VAR model for the bivariate interest rate series. Write down the fitted model.
>
> (b) Compute the impulse response functions of the fitted VAR model. It suffices to use the first 6 lags.
>
> (c) Use the fitted VAR model to produce 1-step- to 12-step-ahead forecasts of the interest rates, assuming that the forecast origin is March 2004.
>
> (d) Are the two interest rate series cointegrated, when a restricted constant term is used? Use 5% significance level to perform the test.
>
> (e) If the series are cointegrated, build an ECM for the series. Write down the fitted model.
>
> (f) Use the fitted ECM to produce 1-step- to 12-step-ahead forecasts of the interest rates, assuming that the forecast origin is March 2004.
>
> (g) Compare the forecasts produced by the VAR model and the ECM.

Solution:

Block 2 — Factor models

(Tsay Ch. 9 • Question 5)

Question 5. A macroeconomic factor model with VAR surprises (Tsay 9.7).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 9, Exercise 9.7, printed p. 503. Reproduced verbatim.

> 9.7. The file `m-fedip.txt` contains year, month, effective federal funds rate, and the industrial production index from July 1954 to December 2003. The industrial production index is seasonally adjusted. Use the federal funds rate and the industrial production index as the macroeconomic variables. Fit a macroeconomic factor model to the 10 excess returns in `m-excess-c10sp-9003.txt`. You can use a VAR model to obtain the surprise series of the macroeconomic variables. Comment on the fitted factor model.

Solution:

Block 3 — Multivariate volatility models

(Tsay Ch. 10 • Questions 6–11)

Common setup for Questions 6–11. Returns are converted to log returns in percent, $r_{it} = 100 \ln(1 + R_{it})$. All models are estimated by Gaussian maximum likelihood, the log density of a shock $\boldsymbol{a}_t$ being $-\frac{1}{2}[k \ln 2\pi + \ln |\boldsymbol{\Sigma}_t| + \boldsymbol{a}_t' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{a}_t]$; the recursions start at the sample covariance; standard errors come from the inverse numerical Hessian. Adequacy uses the multivariate Ljung–Box statistic $Q_k(m)$ of Tsay's Eq. (8.7) applied to the standardized residuals $\tilde{\boldsymbol{a}}_t = \boldsymbol{\Sigma}_t^{-1/2} \boldsymbol{a}_t$ (symmetric square root) and to $\tilde{\boldsymbol{a}}_t^2$, with $k^2 m$ degrees of freedom (Tsay p. 533). Figures plot every sixth month.

Question 6. EWMA covariance matrix for IBM, HPQ and the S&P index (Tsay 10.1).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 10, Exercise 10.1, printed p. 554. Reproduced verbatim.

> 10.1. Consider the monthly simple returns, including dividends, of IBM stock, Hewlett-Packard (HPQ) stock, and the S&P composite index from January 1962 to December 2008 for 564 observations. The returns are in the file `m-ibmhpqsp6208.txt`. Transform into log returns in percentages. Use the exponentially weighted moving-average method to obtain a multivariate volatility series for the three return series. What is the estimated λ? Plot the three volatility series.

Solution:

Question 7. DVEC(1,1) model for IBM and HPQ (Tsay 10.2).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 10, Exercise 10.2, printed p. 554. Reproduced verbatim.

> 10.2. Focus on the monthly log returns of IBM and HPQ stocks from January 1962 to December 2008. Fit a DVEC(1,1) model to the bivariate return series. Is the model adequate? Plot the fitted volatility series and the time-varying correlations.

Solution:

Question 8. BEKK(1,1) model for the S&P index and HPQ (Tsay 10.3).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 10, Exercise 10.3, printed p. 554. Reproduced verbatim.

> 10.3. Focus on the monthly log returns of the S&P composite index and HPQ stock. Build a BEKK model for the bivariate series. What is the fitted model? Plot the fitted volatility series and the time-varying correlations.

Solution:

Question 9. Constant-correlation model for IBM, HPQ and the S&P index (Tsay 10.4).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 10, Exercise 10.4, printed p. 554. Reproduced verbatim.

> 10.4. Build a constant-correlation volatility model for the three monthly log returns of IBM stock, HPQ stock, and S&P composite index. Write down the fitted model. Is the model adequate? Why?

Solution:

Question 10. Constant-correlation GARCH for GE and the S&P index, with a covariance forecast (Tsay 10.5).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 10, Exercise 10.5, printed p. 554. Reproduced verbatim.

> 10.5. The file `m-geibmsp2608.txt` contains the monthly simple returns of General Electric stock, IBM stock, and the S&P composite index from January 1926 to December 2008. The returns include dividends. Transform into log returns in percentages. Focus on the monthly log returns in percentages of GE stock and the S&P 500 index. Build a constant-correlation GARCH model for the bivariate series. Check the adequacy of the fitted model, and obtain the 1-step-ahead forecast of the covariance matrix at the forecast origin December 2008.

Solution:

Question 11. Dynamic conditional correlation model for GE, IBM and the S&P index (Tsay 10.6).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 10, Exercise 10.6, printed p. 554. Reproduced verbatim.

> 10.6. Again, consider the monthly log returns of GE, IBM, and S&P composite index from January 1926 to December 2008. Build a dynamic correlation model for the three-dimensional series. For simplicity, use the sample correlation matrix for $\boldsymbol{\rho}$ in Eq. (10.32).

Solution:

Block 4 — Bayesian SUR with serially correlated errors *(Greenberg Ch. 10 • Question 12)*

Question 12. SUR model with first-order autocorrelation: a Gibbs algorithm (Greenberg 10.4).
Source: Greenberg (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 10 (Time Series), Section 10.6, Exercise 10.4, printed p. 167. Reproduced verbatim.

> 10.4 Modify the SUR model to include first-order autocorrelation, and describe an algorithm for estimating parameters.

Solution: